United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam Spring 2026

CSE 2233/CSI 233: Theory of Computation/Theory of Computing
Total Marks: **40** Duration: 2 Hours
Answer all questions. Figures are in the right-hand margin indicates full marks.
Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

---

1. Consider the following Context-free grammars (CFG) and answer according to it: [4x2]

a) Using a Top-Down Parse Tree, derive the following string: **{$$$##$$}{$#$$}**

S → {A}S | ε
A → B A | B
B → $B# | #B$ | {S} | $

b) With the help of rightmost derivations, find out if the grammar is ambiguous or not for the String: **{ab}ab**

S → SS | {S}| aSb | bSa | a | b | ε

2. Design **CFGs** that generate the following languages: [2x4]

a) L = { $a^{p+2} b^q c^{r+1}$ | $p+q = r+2$ and $p, q, r \geq 1$ }
b) L = { $x^a y^b z^c$ | $a = c$ or $b = 2a$, $a, b, c \geq 0$ }
c) L = { w is a string which does not contain “aa” as a substring, Σ ={a,b} }
d) L = { w is a string, where 1 is present in every odd position, Σ ={0,1,2} }

3. Convert the following grammars into **Chomsky Normal Form (CNF)**: [4x2]

a) S→aAbB | ABC
A→aA | aa
B→bB | ε
C→cC | ε

b) S → aSb | bSa | aA | ε
A → aABB | a
B → b

4. Draw the **Push Down Automata (PDA)** for the following languages: [4x2]

a) L = { $a^n b^m c^k$ | n + 2m = 2k where n, m, k ≥ 1 and n is an even number }
Example: aaaabbbccccc ($a^4b^3c^5$), aabbbcccc ($a^2b^3c^4$)

b) L = { $p^n \# q^m$ | n ≤ m where n, m ≥ 1 }
Example: ppp#qqqqq ($p^3\#q^5$), pp#qq ($p^2\#q^2$)

5. Draw the **Turing Machine (TM)** for the following languages (No need to show tape traversal. You can use it for rough): [4x2]

a) L = { $a^{2x} b^y c^{3y} d^x$ | x, y ≥ 1}
Example: aaaabbccccccdd ($a^4b^2c^6d^2$), aabcccd ($a^2b^1c^3d^1$)

b) L = { $p^{2m} q^n r^k$ | where $n = \frac{k}{m}$ and m, n, k>=1}
Example: ppppqqqrrrrrr($p^4q^3 r^6$), ppqqqrrr($p^2q^3 r^3$)